INNER PRODUCT

The Pythagorean Theorem

Click the hypothesis to see what it does and what breaks without it.

IF

1. $\mathbf{u} \cdot \mathbf{v} = 0$

   CLICK: DROP IT

$\Longrightarrow$

THEN

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

Lengths decompose into independent, additive contributions from each perpendicular direction; the sum may run over any number of pairwise orthogonal vectors.